# Collection round 3: recovered instruments, full-text re-reads and supplementary fields (v5)


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- Supplementary fields: whom a rule binds, when it applies, what a standard's legal effect is, when the robot provision arrived
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*Companion to the analysis report. Round 3 executed the items of the round-3 plan that an agent can do alone: re-fetching blocked and missing instruments by official alternative routes, a harvest with service-robot terms, a full-text re-read of truncated statutes, a supplementary schema (whom an instrument binds, dates, legal effect of standards, provision dating), a second precision audit, and a source split of the governance-gap index. Items that need a person are listed in HUMAN_ASSISTANCE.md.*

## Summary


Round 3 added 160 usable documents to the collection: 108 round-2 candidates whose text could now be read (80 of them principal binding embodied instruments) and 52 new documents, 9 of them reference-list instruments fetched from their titles and the rest from the harvest with service-robot, care, exoskeleton, humanoid and collaborative-robot terms and EuroVoc concepts. The harvest itself added little: of its 43 usable documents only 8 fall in the embodied set and 8 are principal instruments, so the open legal databases hold few service-robot instruments that rounds 1 and 2 had not already found, and the gains of round 3 come from reading texts that were already known. Of the reference-list instruments, 4 are now principal embodied instruments in the collection. The collection now holds 3,320 usable documents, 1,704 in the embodied set, 598 principal embodied instruments and 278 principal binding ones (v4: 3,165 / 1,603 / 505 / 195). Of the 206 round-3 targets, the collectors saved a text for 121, found 74 blocked at official hosts, judged 9 irrelevant and could not locate 2 (Table 80).

The collectors' own reports, with the alternative routes that worked (Publications Office CELEX resolver for EUR-Lex, state legislatures' PDF endpoints, Riigi Teataja and legislation.govt.nz rendered pages, the Kremlin acts bank, Gazzetta Ufficiale issue PDFs) and the metadata corrections they noticed (act numbers, repeal dates, mislabelled gazette copies), are in collection_round3/status/report_*.md.


*Table 80. Round-3 collector outcomes for the 206 targets (189 verified-but-unread binding candidates from round 2 and 17 reference-list instruments missing on the strict test)*

| outcome | targets |
|---|---|
| saved | 113 |
| blocked | 74 |
| irrelevant | 9 |
| saved_partial | 8 |
| not_found | 2 |
| not attempted | 0 |

The blocked targets are the manual-download list for a person with ordinary browser access (Table 81).

*Table 81. Targets whose official hosts refuse the identified research crawler: the list for manual download*

| candidate_id | title_en | jurisdiction | source_url | not |
|---|---|---|---|---|
| EN-014 | Ordinance No. 2016-1057 of 3 August 2016 on the experimentation of vehicles with delegated driving on public roads | FR | https://www.legifrance.gouv.fr/eli/ordonnance/2016/8/3/2016-1057/jo/texte | leg oth |
| EN-015 | Decree No. 2018-211 of 28 March 2018 on the experimentation of vehicles with delegated driving on public roads | FR | https://www.legifrance.gouv.fr/eli/decret/2018/3/28/2018-211/jo/texte | leg oth |
| EN-016 | Law No. 2019-1428 of 24 December 2019 on mobility orientation (LOM) – articles 31–32 on autonomous and connected vehicles | FR | https://www.legifrance.gouv.fr/jorf/id/JORFTEXT000039666574 | leg oth |

| candidate_id | title_en | jurisdiction | source_url | no |
|---|---|---|---|---|
| EN-018 | Decree No. 2021-873 of 29 June 2021 implementing Ordinance No. 2021-443 on criminal liability and conditions of use of vehicles with delegated driving | FR | https://www.legifrance.gouv.fr/eli/decret/2021/6/29/2021-873/jo/texte | leg oth |
| EN-019 | Law No. 2016-1428 of 24 October 2016 on strengthening the safety of civil drone use | FR | https://www.legifrance.gouv.fr/jorf/id/JORFTEXT000033293745 | leg oth |
| EN-038 | Royal Decree of 10 April 2016 on the use of remotely piloted aircraft in Belgian airspace | BE | https://www.ejustice.just.fgov.be/eli/arrete/2016/04/10/2016014132/justel | Th nu Mo the del tel |
| EN-050 | Law No. 2015-992 of 17 August 2015 on energy transition for green growth – article 37 IX (authorisation to legislate by ordinance on vehicles with delegated driving) | FR | https://www.legifrance.gouv.fr/loda/id/JORFTEXT000031044385/ | leg oth 20 |
| EN-051 | Order of 17 April 2018 on the experimentation of vehicles with delegated driving on public roads | FR | https://www.legifrance.gouv.fr/loda/id/JORFTEXT000036868691/ | leg oth |
| EN-052 | Law No. 2019-486 of 22 May 2019 on business growth and transformation (PACTE) – article 125 on automated vehicle experimentation | FR | https://www.legifrance.gouv.fr/jorf/article_jo/JORFARTI000038496320 | leg oth |
| EN-053 | Ordinance No. 2021-442 of 14 April 2021 on access to vehicle data | FR | https://www.legifrance.gouv.fr/loda/id/JORFTEXT000043370884 | leg oth |
| EN-054 | Decree No. 2022-1034 of 21 July 2022 publishing the amendment to the 1968 Vienna Convention on Road Traffic (automated driving systems) | FR | https://www.legifrance.gouv.fr/jorf/id/JORFTEXT000046081252 | leg oth Vie |
| EN-055 | Order of 2 August 2022 on the qualification of remote operators of automated road transport systems (art. R. 3152-3 Code des transports) | FR | https://www.legifrance.gouv.fr/jorf/id/JORFTEXT000046151685 | leg oth |
| EN-056 | Order of 2 August 2022 on the approval procedure for qualified bodies assessing automated road transport systems (art. R. 3152-30 Code des transports) | FR | https://www.legifrance.gouv.fr/jorf/id/JORFTEXT000046216833 | leg oth |

| candidate_id | title_en | jurisdiction | source_url | not |
|---|---|---|---|---|
| EN-057 | Order of 8 December 2022 on the mandatory consumer information about conditions of use of a vehicle's automated driving system | FR | https://www.legifrance.gouv.fr/jorf/id/JORFTEXT000046828949 | leg oth |
| EN-058 | Decree No. 2023-644 of 20 July 2023 on access to certain vehicle data for accident prevention, road infrastructure mapping and traffic knowledge | FR | https://www.legifrance.gouv.fr/jorf/id/JORFTEXT000047867363 | leg oth |
| EN-059 | Decree No. 2024-1063 of 25 November 2024 on automated road freight transport | FR | https://www.legifrance.gouv.fr/jorf/id/JORFTEXT000050662872 | leg oth |
| EN-060 | Order of 15 April 2025 on operational safety of automated road transport systems using fully automated vehicles for public passenger transport | FR | https://www.legifrance.gouv.fr/jorf/id/JORFTEXT000051484894 | leg oth |
| EN-084 | Decree-Law No. 58/2018 of 23 July – registration and compulsory civil liability insurance regime for unmanned aircraft (drones) | PT | https://diariodarepublica.pt/dr/detalhe/decreto-lei/58-2018-115898563 | dia file op 20 |
| EN-085 | ANAC Regulation No. 1093/2016 of 14 December – operating conditions for use of airspace by civil remotely piloted aircraft systems (drones) | PT | https://diariodarepublica.pt/dr/detalhe/regulamento/1093-2016-105367104 | dia file op bot |
| EU-130 | Act on the testing of self-driving vehicles (Norway) | NO | https://lovdata.no/dokument/NL/lov/2017-12-15-112 | lov luft sh |
| EU-131 | Regulation on the testing of self-driving motor vehicles (Norway) | NO | https://lovdata.no/dokument/SF/forskrift/2017-12-19-2240 | lov luft sh |
| EU-132 | Regulation on unmanned aircraft (Norway, BSL A 7-2) | NO | https://lovdata.no/dokument/SF/forskrift/2024-11-01-2777 | lov luft sh |
| EU-133 | Regulation on aircraft without a pilot on board etc. (Norway, 2015) | NO | https://lovdata.no/dokument/SF/forskrift/2015-11-30-1404 | lov luft sh |
| EU-134 | Regulation on U-space and an integrated digital aeronautical publication (Norway, BSL A 7-3) | NO | https://lovdata.no/dokument/SF/forskrift/2024-10-23-2572 | lov luft sh |
| AP-002 | Drone (Amendment) Rules, 2022 (G.S.R. 108(E)) | IN | https://www.civilaviation.gov.in/sites/default/files/2023-10/Drone%20%28Amendment%29%20Rules%2C%202022%20dated%2011%20Feb%202022_0.pdf | civ an blo htt dat |
| AP-009 | Conditional exemption to Government of Telangana for drone operations (MoCA order, 4 August 2021) | IN | https://www.civilaviation.gov.in/node/3286 | civ to ww exe arc do |

| candidate_id | title_en | jurisdiction | source_url | not |
|---|---|---|---|---|
| AP-087 | Motor Vehicles (Trials of Automotive Technologies) Amendment Act 2016 (SA) | AU | https://www.legislation.sa.gov.au/_legislation-documents/lz/v/a/2016/motor-vehicles-trials-of-new-automotive-technologies-amendment-act-2016_10/2016.10.un.pdf | leg an 20 exi |
| AP-088 | Transport Legislation Amendment (Automated Vehicle Trials and Innovation) Act 2017 No 22 (NSW) | AU | https://legislation.nsw.gov.au/view/pdf/asmade/act-2017-41 | leg PD Ho Au Tra |
| AP-089 | Road Safety Amendment (Automated Vehicles) Act 2018 (Vic) | AU | https://www.legislation.vic.gov.au/as-made/acts/road-safety-amendment-automated-vehicles-act-2018 | leg co link A h |
| EN-086 | Turkish DGCA Instruction on Unmanned Aircraft Systems (SHT-UAV) and UAV registration system | TR | https://web.shgm.gov.tr/documents/sivilhavacilik/files/mevzuat/SHT-IHA_Rev-05.pdf | TL: att po htt pa (re me |
| IG-112 | DoD Directive 3000.09, Autonomy in Weapon Systems (updated 25 January 2023) | US | https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodd/300009p.pdf | es oth |
| ML-032 | Federal Decree-Law No. (26) of 2022 Concerning the Regulation of Unmanned Aircraft | AE | https://uaelegislation.gov.ae/en/legislations/1587 | ua /d is |
| ML-033 | Cabinet Resolution No. (110) of 2023 Concerning the UAE Scheme for Drone Products and Systems (UAE Regulation for Unmanned Aircraft) | AE | https://uaelegislation.gov.ae/en/legislations/2177 | sa |
| ML-041 | GACAR Part 107 - Operation of Unmanned Aircraft Systems (Version 4) | SA | https://gaca.gov.sa/-/media/Files/PDF/LawsAndRegulation/Aviation-Safety-and-Environmental-Sustainability/GACAR-Safety-Regulations/107-v4.pdf | ww sa htt En sm |
| ML-096 | DAN 151 - Operations of remotely piloted aircraft (RPAS) in matters of public interest | CL | https://www.dgac.gob.cl/ | dg (ht am sol 20 fro |
| US-086 | California Vehicle Code Section 38750: Autonomous vehicles (SB 1298, 2012) | US | https://leginfo.legislature.ca.gov/faces/codes_displaySection.xhtml?lawCode=VEH&sectionNum=38750. | leg twi sa |
| US-108 | Indiana Code 9-21-11.5: Personal Delivery Devices (HEA 1072) | US | https://iga.in.gov/laws/2024/ic/titles/9#9-21-11.5 | iga Hu |
| US-109 | Indiana Code 35-33-5-9: Warrant requirement for unmanned aerial vehicles (HEA 1009, 2014) | US | https://iga.in.gov/laws/2024/ic/titles/35#35-33-5-9 | iga |
| US-122 | Michigan SAVE Act package (Public Acts 332-335 of 2016; MCL 257.665 et seq.): Automated motor vehicles | US | https://www.legislature.mi.gov/Laws/MCL?objectName=mcl-257-665 | leg an |
| US-123 | Michigan Unmanned Aircraft Systems Act (Public Act 436 of 2016; MCL 259.301-259.335) | US | https://www.legislature.mi.gov/Laws/MCL?objectName=mcl-Act-436-of-2016 | leg |

| candidate_id | title_en | jurisdiction | source_url | no |
|---|---|---|---|---|
| US-134 | New Hampshire RSA 265:163: Personal delivery devices and mobile carriers (HB 116, 2022) | US | https://gc.nh.gov/rsa/html/XXI/265/265-163.htm | gc. (M |
| US-182 | Wisconsin Executive Order #245 (2017): Governor's Steering Committee on Autonomous and Connected Vehicle Testing and Deployment | US | https://docs.legis.wisconsin.gov/code/executive_orders/2011_scott_walker/2017-245.pdf | do the |
| REF-06 | ISO/TS 15066:2016 collaborative robots | INT | https://www.iso.org/standard/62996.html | iso hu |
| REF-08 | ISO 26262 road vehicle functional safety | INT | https://www.iso.org/standard/68383.html | iso |
| REF-09 | ISO 21448 (SOTIF) | INT | https://www.iso.org/standard/77490.html | iso |
| REF-12 | ANSI/RIA R15.06 industrial robot safety | US | https://webstore.ansi.org/standards/ria/ansiriar15062012 | a3 AN |
| REF-13 | ANSI/A3 R15.08 industrial mobile robots | US | https://webstore.ansi.org/standards/ria/ansia3r15082020 | a3 |
| AP-020 | Unmanned Aircraft (Public Safety and Security) Act 2015 (Act 16 of 2015) | SG | https://sso.agc.gov.sg/Acts-Supp/16-2015 | ss |
| AP-036 | CAAT Announcement No. 3 B.E. 2568 (2025) on rules for agricultural drone operations in border/security areas | TH | https://www.caat.or.th/caat-media/167323/ | ca po an ap |
| AS-015 | Tianjin Municipal Regulations on Promoting the Development of Intelligent Connected Vehicles | CN | https://gyxxh.tj.gov.cn/ZWGK4147/ZCWJ6355/sjwj/202512/t20251204_7191311.html | tj.g Alt |
| AS-016 | Yangquan Municipal Measures for the Administration of Intelligent Connected Vehicles | CN | https://flk.npc.gov.cn/detail?id=ff8081818c24df58018cce7f027466b2 | flk. co |
| AS-017 | Hefei Municipal Regulations on Promoting the Application of Intelligent Connected Vehicles | CN | http://www.ahrd.gov.cn/article.jsp?strId=7a876823fe4544c1a9850923d460665a&strColId=44861419b53244f8910cd00be098270d&strWebSiteId=1448865560847002 | An fro |
| AS-018 | Wuhan Municipal Regulations on Promoting the Development of Intelligent Connected Vehicles | CN | https://flk.npc.gov.cn/detail?id=ff808181927f1276019487a8b5db616c | flk res tex |
| AS-019 | Guangzhou Municipal Regulations on Innovative Development of Intelligent Connected Vehicles | CN | https://flk.npc.gov.cn/detail?id=ff808181927f0e7b0194de2edeb60a92 | flk |

| candidate_id | title_en | jurisdiction | source_url | no |
|---|---|---|---|---|
| AS-020 | Ordos Municipal Regulations on Promoting the Development of Intelligent Connected Vehicles | CN | https://flk.npc.gov.cn/detail?id=ff8081819aedd63e019b495526357047 | flk co the |
| AS-023 | Changsha Municipal Regulations on the Development of Autonomous Vehicles | CN | https://flk.npc.gov.cn/detail?id=ff8081819bdff392019be3364efd0ab4 | flk dat |
| AS-026 | Huzhou Municipal Regulations on Innovative Development of the Intelligent Connected Vehicle Industry | CN | https://flk.npc.gov.cn/detail?id=ff8081819f12cdb7019f3bae7cbf4adc | flk |
| AS-027 | Jiaxing Municipal Regulations on Promoting Road Testing and Application of Intelligent Connected Vehicles | CN | https://flk.npc.gov.cn/detail?id=ff8081819dba1254019dce0118552f83 | flk |
| AS-031 | Guangzhou Municipal Regulations on the Development of the Low-Altitude Economy | CN | https://ghzyj.gz.gov.cn/zwgk/newzcfg/ywly/content/post_10120201.html | gz. htt |
| AS-032 | Zhuhai Special Economic Zone Regulations on Low-Altitude Transport Construction and Management | CN | https://flk.npc.gov.cn/detail?id=ff808181927f093101939b3d540500a6 | flk |
| AS-036 | Zhejiang Provincial Provisions on Public Security Administration of Unmanned Aircraft | CN | https://flk.npc.gov.cn/detail?id=2c90e5ba6a291d8f016a4cc775c3034a | flk |
| AS-038 | Zhejiang Provincial Civil Aviation Regulations | CN | https://flk.npc.gov.cn/detail?id=ff8081819f12cdb7019f369a9f932c52 | flk |
| AS-104 | Enforcement Rule of the Autonomous Vehicles Act | KR | https://www.law.go.kr/LSW/lsInfoP.do?lsiSeq=271493 | law |
| AS-106 | Rules on Performance and Standards of Motor Vehicles and Parts (Motor Vehicle Safety Standards) | KR | https://www.law.go.kr/LSW/lsInfoP.do?lsiSeq=288083 | law |
| AS-112 | Enforcement Rule of the Aviation Safety Act | KR | https://www.law.go.kr/LSW/lsInfoP.do?lsiSeq=287951 | law an |
| AS-114 | Act on Promotion of Development and Commercialization of Autonomous Ships | KR | https://www.law.go.kr/LSW/lsInfoP.do?lsiSeq=281889 | law (en |
| AS-115 | Rules on Occupational Safety and Health Standards | KR | https://www.law.go.kr/LSW/lsInfoP.do?lsiSeq=273603 | law rul |
| AS-121 | Regulations on Issuance of Plates for Unmanned Vehicle Innovative Experimentation Projects | TW | https://law.moj.gov.tw/LawClass/LawAll.aspx?pcode=K0090084 | law mo |
| AS-123 | Civil Aviation Act (Chapter 9-2 on Remotely Piloted Aircraft) | TW | https://law.moj.gov.tw/LawClass/LawAll.aspx?pcode=K0090001 | law a= |

| candidate_id | title_en | jurisdiction | source_url | no |
|---|---|---|---|---|
| AS-125 | Rules for Investigation of Major Flight Accidents Involving Remotely Piloted Aircraft | TW | https://law.moj.gov.tw/LawClass/LawAll.aspx?pcode=K0090086 | law |
| AS-126 | Regulations on Delegation of Drone Inspection and Operator Testing | TW | https://law.moj.gov.tw/LawClass/LawAll.aspx?pcode=K0090085 | law |
| AS-127 | Regulations on Application and Management of Drone Flights within Designated Commercial Port Areas | TW | https://law.moj.gov.tw/LawClass/LawAll.aspx?pcode=K0080041 | law |
| AS-134 | Notice of Beijing Municipal Government Publishing the Scope of Controlled Airspace for Unmanned Aircraft | CN | https://www.beijing.gov.cn/zhengce/zhengcefagui/202508/t20250804_4165802.html | No (W (ab in t |
| REF-16 | Taiwan Industrial Robot Hazard Prevention Standards (1987, amended 2018) | TW | https://law.moj.gov.tw/LawClass/LawSearchContent.aspx?kw=工業用機器人危害預防標準 | law En |

## Full-text re-reads of truncated instruments

The excerpt policy stops at 110,000 characters; 104 principal embodied instruments exceeded it. Re-reading them with up to 400,000 characters under the same prompt left the document class unchanged in 96.2% and the relevance rating in 87.5%; principle levels agreed exactly in 89.7% of field-document pairs, and the full text raised a level in 6.4% of pairs (Table 74). The full-text labels replace the excerpt labels for these documents. The documents whose set-defining labels changed are listed in Table 75.

*Table 74. Full-text re-read (up to 400,000 characters) against the excerpt-based label (110,000 characters) for the 104 principal instruments whose text exceeded the excerpt: agreement per field*

| field | n | agree | agree_pct |
|---|---|---|---|
| doc_class | 104 | 100 | 96.2 |
| binding_nature | 104 | 101 | 97.1 |
| instrument_status | 104 | 100 | 96.2 |
| ai_scope | 104 | 93 | 89.4 |
| embodiment_status | 104 | 96 | 92.3 |
| relevance_to_embodied_ai_governance | 104 | 91 | 87.5 |
| page_quality | 104 | 96 | 92.3 |
| primary_embodied_category | 104 | 93 | 89.4 |
| risk_framing | 104 | 90 | 86.5 |
| embodied_categories (mean Jaccard) | 104 | 88.7 | 85.3 |
| governance_topics (mean Jaccard) | 104 | 79.8 | 76.7 |
| governance_mechanisms (mean Jaccard) | 104 | 82.6 | 79.4 |
| application_sectors (mean Jaccard) | 104 | 89.4 | 86 |
| principles (exact level, all 20) | 2080 | 1865 | 89.7 |
| principle levels raised by the full text | 2080 | 134 | 6.4 |
| principle levels lowered by the full text | 2080 | 81 | 3.9 |

*Table 75. Truncated principal instruments whose set-defining labels changed when the full text was read*

| doc_id | title | changed_fields | excerpt_class | fulltext_class | excerpt_relevance | fulltext_relevance |
|---|---|---|---|---|---|---|
| R2-7bb2229f | Regulation (EU) 2023/1230 of the European Parliament and of the Council of 14 June 2023 on machinery and repealing Directive 2006/42/EC and Council Directive 73/361/EEC | ai_scope | governance_instrument_binding | governance_instrument_binding | 3 | 3 |
| R2-8991668e | Official Mexican Standard NOM-107-SCT3-2019, establishing requirements to operate a remotely piloted aircraft system (RPAS) in Mexican airspace | relevance_to_embodied_ai_governance | governance_instrument_binding | governance_instrument_binding | 3 | 2 |
| R2-6da0ddec | Royal Decree 517/2024 of 4 June developing the legal regime for the civil use of unmanned aircraft systems (UAS) | relevance_to_embodied_ai_governance | governance_instrument_binding | governance_instrument_binding | 2 | 3 |
| R2-82f8813a | Civil Aviation Regulations 2016 | ai_scope | governance_instrument_binding | governance_instrument_binding | 1 | 1 |
| R2-f67080c2 | Standards Document: Operation of Aircraft, Part IV – International Operations – Remotely Piloted Aircraft Systems (RPAS) | relevance_to_embodied_ai_governance | governance_instrument_binding | governance_instrument_binding | 3 | 2 |
| D1315 | White Paper of the Austrian Council on Robotics and Artificial Intelligence | relevance_to_embodied_ai_governance | governance_proposal_or_draft | governance_proposal_or_draft | 2 | 3 |
| R2-c5eb5314 | Civil Aviation Safety Regulations 1998 | ai_scope | governance_instrument_binding | governance_instrument_binding | 2 | 2 |
| D1342 | National AI Strategy | ai_scope | governance_instrument_soft | governance_instrument_soft | 1 | 1 |

| doc_id | title | changed_fields | excerpt_class | fulltext_class | excerpt_relevance | fulltext_relevance |
|---|---|---|---|---|---|---|
| D1290 | Proposal for a Regulation of the European Parliament and of the Council Laying Down Harmonised Rules on Artificial Intelligence (Artificial Intelligence Act) and Amending Certain Union Legislative Acts | binding_nature | governance_proposal_or_draft | governance_proposal_or_draft | 2 | 2 |
| R2-f57251ed | Ordinance on Industrial Safety and Health | relevance_to_embodied_ai_governance | governance_instrument_binding | governance_instrument_binding | 2 | 1 |
| D1374 | Proposals for an Italian Strategy for Artificial Intelligence | doc_class; relevance_to_embodied_ai_governance | governance_instrument_soft | governance_proposal_or_draft | 1 | 2 |
| R2-6ecde273 | Decision No. 2021-817 DC of 20 May 2021 - Global Security Law Preserving Freedoms | relevance_to_embodied_ai_governance | governance_instrument_binding | governance_instrument_binding | 2 | 1 |
| R2-f48a8103 | Proposal for a Regulation of the European Parliament and of the Council on machinery products | binding_nature; ai_scope | governance_proposal_or_draft | governance_proposal_or_draft | 3 | 3 |
| R2-9c4dc434 | Civil Aviation Law | ai_scope | governance_instrument_binding | governance_instrument_binding | 1 | 1 |
| R2-1617d407 | Civil Aviation Directive 6011 Part (V) – Unmanned Aircraft System: Special UAS Project | relevance_to_embodied_ai_governance | governance_instrument_binding | governance_instrument_binding | 3 | 2 |
| D1186 | Hungary's Artificial Intelligence Strategy 2020–2030 | relevance_to_embodied_ai_governance | governance_instrument_soft | governance_instrument_soft | 1 | 2 |
| R2-7115e06c | Opinion on the Integration of Autonomy into Lethal Weapon Systems | doc_class | governance_instrument_soft | government_report_or_parliamentary | 3 | 3 |

| doc_id | title | changed_fields | excerpt_class | fulltext_class | excerpt_relevance | fulltext_relevance |
|---|---|---|---|---|---|---|
| R2-92b0acef | Proposal for a Regulation amending Regulations (EU) 2017/745 and (EU) 2017/746 as regards simplifying and reducing the burden of the rules on medical devices and in vitro diagnostic medical devices, and amending Regulations (EU) 2022/123 and (EU) 2024/1689 | binding_nature; relevance_to_embodied_ai_governance | governance_proposal_or_draft | governance_proposal_or_draft | 2 | 1 |
| R2-2b6aea0b | Ordinance on Type Designation of Devices (Road Vehicles) | ai_scope | governance_instrument_binding | governance_instrument_binding | 2 | 2 |
| R2-b70da4e3 | Our Common Agenda Policy Brief 9: A New Agenda for Peace | doc_class; relevance_to_embodied_ai_governance | governance_instrument_soft | governance_proposal_or_draft | 2 | 1 |
| R2-80171de4 | Directive 2006/42/EC of the European Parliament and of the Council of 17 May 2006 on machinery, and amending Directive 95/16/EC (recast) | ai_scope | governance_instrument_binding | governance_instrument_binding | 2 | 2 |
| R2-a586cca8 | Regulation No 79 of the Economic Commission for Europe of the United Nations (UN/ECE) — Uniform provisions concerning the approval of vehicles with regard to steering equipment | relevance_to_embodied_ai_governance | governance_instrument_binding | governance_instrument_binding | 2 | 3 |
| R2-35e40c62 | Adra Strategic Research, Innovation, and Deployment Agenda 2025-2027 | doc_class | governance_proposal_or_draft | governance_instrument_soft | 2 | 2 |
| R2-431b7f3a | Ministerial Ordinance Establishing Technical Standards for Railways | ai_scope | governance_instrument_binding | governance_instrument_binding | 1 | 1 |

| doc_id | title | changed_fields | excerpt_class | fulltext_class | excerpt_relevance | fulltext_relevance |
|---|---|---|---|---|---|---|
| R2-322c6200 | The Workplace Safety and Health Act — Workplace Safety and Health Regulation, Manitoba Regulation 217/2006 | relevance_to_embodied_ai_governance; ai_scope | governance_instrument_binding | governance_instrument_binding | 2 | 1 |
| D0104 | Ethics and governance of artificial intelligence for health: WHO guidance | ai_scope | governance_instrument_soft | governance_instrument_soft | 1 | 1 |

## Supplementary fields: whom a rule binds, when it applies, what a standard's legal effect is, when the robot provision arrived

A second, smaller annotation pass asked five questions the codebook had not asked, of every principal embodied instrument, with a quote required for every answer. Among the 278 principal binding instruments the text names whom the rule binds in 267 cases (operators or deployers and public bodies most often; Table 71), an entry-into-force date in 181, a later application date in 49 and a repeal or expiry date in 15 (Table 70). The year the robot-relevant provision arrived could be given for 226 binding instruments, but mostly because the whole instrument is about such systems (182 cases), so the provision year is the document year; an amendment history in the text dated the provision in 26 cases and the provision itself stated its date in 16; in 54 the year could not be determined from the captured text. Only 7 binding instruments carry a provision year later than the document's recorded year (Table 73). The consolidated statutes dated before 2010 are the case that matters for trend analyses, and for 15 of the 16 the captured consolidation carries no amendment history, so the model could not date their automated-driving or drone articles: these need a person with the amendment record (HUMAN_ASSISTANCE.md, the legal-check item). Among the technical standards, 45 state on their public page that they are voluntary and 1 that they are harmonised or designated under a law (Table 72).

*Table 70. Supplementary fields (v5) stated in the text: counts of principal instruments for which the model found each field in the document, and the number whose robot-relevant provision is dated later than the document's own year*

| population | n | bound_parties_stated | entry_into_force_date | applicability_date | repeal_or_expiry_date | robot_provision_year | provision_year_later_than_document_year |
|---|---|---|---|---|---|---|---|
| principal embodied instruments | 598 | 436 | 287 | 64 | 36 | 526 | 8 |
| principal binding embodied instruments | 278 | 267 | 181 | 49 | 15 | 226 | 7 |
| principal binding, dated before 2010 | 16 | 16 | 9 | 2 | 0 | 1 | 1 |

*Table 71. Whom the 278 principal binding embodied instruments bind, as stated in their text (a document may bind several parties)*

| party | binding_instruments | share_pct |
|---|---|---|
| public_bodies_or_government | 176 | 63.3 |
| operators_or_deployers | 171 | 61.5 |
| manufacturers_or_producers | 108 | 38.8 |
| owners_or_users | 106 | 38.1 |
| pilots_or_drivers | 70 | 25.2 |
| member_states_or_subnational_governments | 68 | 24.5 |
| individuals_or_general_public | 51 | 18.3 |
| certification_or_notified_bodies | 39 | 14 |
| importers_or_distributors | 38 | 13.7 |
| providers_or_developers | 30 | 10.8 |
| employers | 16 | 5.8 |
| none_stated | 9 | 3.2 |
| researchers_or_institutions | 7 | 2.5 |
| not_applicable | 2 | 0.7 |

*Table 72. Legal effect of the 61 principal instruments annotated as technical standards, as stated on the captured text*

| legal_effect | standards | share_pct |
|---|---|---|
| voluntary | 45 | 73.8 |
| unknown | 15 | 24.6 |
| harmonised_or_designated | 1 | 1.6 |

*Table 73. Principal binding embodied instruments whose robot-relevant provision was added later than the document's recorded year, according to the amendment history in the text (the dates to use for provision-level trend analyses)*

| doc_id | title_en | jurisdiction_country | year_final | robot_provision_year | robot_provision_basis |
|---|---|---|---|---|---|
| R2-aa8d88b9 | Ordinance on Transport Safety in the Freight Motor Vehicle Transport Business | JP | 1990 | 2025 | amendment_history_in_text |
| R2-ee56555b | Traffic Act (Estonia) | EE | 2010 | 2017 | amendment_history_in_text |
| R2-5d2b19e5 | Finnish Aviation Act 864/2014 (consolidated version) | FI | 2014 | 2020 | amendment_history_in_text |
| R2-ce2171ab | Ordinance for Enforcement of the National Strategic Special Zones Act (MLIT-related) | JP | 2014 | 2022 | amendment_history_in_text |
| R2-f8592e32 | The Air Navigation Order 2016 | GB | 2016 | 2025 | amendment_history_in_text |
| R2-295bf91b | Air Navigation (Aircraft Noise) Regulations 2018 | AU | 2018 | 2021 | amendment_history_in_text |
| R2-be225994 | Regulations Amending the Canadian Aviation Regulations (Remotely Piloted Aircraft Systems): SOR/2019-11 | CA | 2018 | 2019 | whole_instrument_is_about_such_systems |

## Second precision audit

An independent agent re-read a random sample of 100 of the 195 principal binding embodied instruments of v4 (seed 2027) against the five criteria of the v3 audit. Shares: distinct instrument 85%, binding 99%, substantive embodied provisions 76%, autonomy or AI content 73%, explicit AI 9%; all three main criteria jointly 73% (Wilson 64 to 81), implying about 142 genuine instruments (124 to 157) in the set of 195. The v3 audit of the earlier set of 299 found 54% (44 to 63). The sample was drawn from the v4 set before round 3; applied to the v5 set of 278 principal binding instruments the same share implies about 203 (177 to 224). The failures are mostly binding instruments without a substantive embodied provision (a drone or robot named only in a list of functions or definitions), then register, tracking and catalogue pages and date-only amendments that the counting-unit rules let through. The audit report with the failing documents is review/v5_audit/precision_audit_v5.md.

## The governance gap, split by collection source

Because round 2 searched legal databases for drones, vehicles and machinery, the instrument side of the governance-gap index could inherit the search terms. Table 7b computes the index separately against the round-1 instruments (web search of robot-specific and general-AI sources) and against the round-2 and round-3 instruments (database harvests and recovered texts). The gap is positive against both sources for social companion robots (+35 and +41 points), humanoid general purpose (+6 and +20 points), educational robots (+5 and +6 points), sex robots (+3 and +4 points), other embodied (+3 and +4 points): research attention to these machine classes exceeds their share of instruments whichever way the instruments were found, so the headline gap for service, companion and humanoid robots is not an artefact of the round-2 terms. The instrument surplus is present against both sources for drones uas (-31 and -38), autonomous road vehicles (-55 and -26), mobile industrial logistics (-16 and -5), agricultural robots (-18 and -3), autonomous maritime or rail (-6 and -4), rehabilitation exoskeleton prosthetic (-2 and -2). The gap appears against only one source for care assistive robots (+0 against round-1 instruments, +21 against harvested ones): for these classes the round-1 robot-specific sources supplied standards and guidance that the database harvests did not, so the size of the gap depends on how instruments were found. The sign flips for surgical medical robots (-8 against round-1 instruments, +10 against harvested ones), professional service robots (-15 against round-1 instruments, +10 against harvested ones), collaborative robots (-16 against round-1 instruments, +3 against harvested ones), industrial manipulators (-41 against round-1 instruments, +3 against harvested ones): the round-1 instrument set, drawn from robot-specific sources, is rich in standards for these classes, and the harvested set is not, so for them the gap depends on the source and should not be read as a finding.

*Table 07b. Governance-gap index by collection source: research share (n=509) minus the principal-instrument share among round-1 instruments (web search, n=59) and among round-2 instruments (database harvest, n=539); a gap present in both columns does not depend on the round-2 search terms*

| category | level1 | research_pct | n_research | instrument_pct_round1 | n_instruments_round1 | gap_round1 | instrument_pct_round2 | n_instruments_ro |
|---|---|---|---|---|---|---|---|---|
| social_companion_robots | Service & social | 43 | 219 | 8.5 | 5 | 34.6 | 2.4 | |
| care_assistive_robots | Service & social | 27.5 | 140 | 27.1 | 16 | 0.4 | 6.5 | |
| humanoid_general_purpose | Humanoid & general-purpose | 22.8 | 116 | 16.9 | 10 | 5.8 | 2.4 | |
| surgical_medical_robots | Medical & body-integrated | 14.5 | 74 | 22 | 13 | -7.5 | 4.1 | |
| professional_service_robots | Service & social | 17.5 | 89 | 32.2 | 19 | -14.7 | 7.4 | |
| educational_robots | Service & social | 6.9 | 35 | 1.7 | 1 | 5.2 | 1.1 | |
| domestic_household_robots | Service & social | 7.7 | 39 | 8.5 | 5 | -0.8 | 2.8 | |
| sex_robots | Service & social | 4.5 | 23 | 1.7 | 1 | 2.8 | 0.6 | |
| other_embodied | Other embodied | 4.7 | 24 | 1.7 | 1 | 3 | 1.1 | |

| category | level1 | research_pct | n_research | instrument_pct_round1 | n_instruments_round1 | gap_round1 | instrument_pct_round2 | n_instruments_ro |
|---|---|---|---|---|---|---|---|---|
| biohybrid_or_cyborg | Medical & body-integrated | 2.2 | 11 | 0 | 0 | 2.2 | 0.4 | |
| collaborative_robots | Industrial & logistics | 6.3 | 32 | 22 | 13 | -15.7 | 3.2 | |
| toys_entertainment_robots | Service & social | 1.2 | 6 | 0 | 0 | 1.2 | 0.6 | |
| military_autonomous_weapons | Security & military | 10.4 | 53 | 20.3 | 12 | -9.9 | 9.1 | |
| space_or_extreme_environment | Vehicles & field | 3.1 | 16 | 10.2 | 6 | -7 | 2.6 | |
| industrial_manipulators | Industrial & logistics | 10 | 51 | 50.8 | 30 | -40.8 | 7.2 | |
| security_policing_robots | Security & military | 3.1 | 16 | 6.8 | 4 | -3.6 | 4.5 | |
| rehabilitation_exoskeleton_prosthetic | Medical & body-integrated | 2.8 | 14 | 5.1 | 3 | -2.3 | 5 | |
| autonomous_maritime_or_rail | Vehicles & field | 1 | 5 | 6.8 | 4 | -5.8 | 4.6 | |
| agricultural_robots | Vehicles & field | 2.4 | 12 | 20.3 | 12 | -18 | 5.4 | |
| mobile_industrial_logistics | Industrial & logistics | 3.1 | 16 | 18.6 | 11 | -15.5 | 7.8 | |
| delivery_robots | Industrial & logistics | 2.2 | 11 | 3.4 | 2 | -1.2 | 9.6 | |
| autonomous_road_vehicles | Vehicles & field | 12.4 | 63 | 67.8 | 40 | -55.4 | 38.4 | |
| drones_uas | Vehicles & field | 6.3 | 32 | 37.3 | 22 | -31 | 43.8 | |

## What round 3 did not do

No person labelled any document (the human-validation kit is in data/human_validation/); hosts that refuse the identified crawler were not circumvented (Table 81 lists them for manual download); technical standards are still read from scope pages; and the labels of the new and recovered documents carry the same model limits as the rest.